

École Pour l'Informatique et les Techniques Avancées – EPITA

Masters program

Course: Introduction to Network protocols and Architecture (INPA)

INPA Course Outline

Course schedule (tentative)

Date	No.	Topics	Duration (in hours)
(check Zeus platform for the date & time)	1	Primer, TCP/IP stack, TCP/IP stack operations, Class exercises	3
(check Zeus platform for the date & time)	2	IP address allocation & routing, Subnetting, ARP, Class exercises	3
(check Zeus platform for the date & time)	3	DHCP, Packet handling, Routing Protocols (RIP, OSPF), Class exercises	3
(check Zeus platform for the date & time)	4	TCP, UDP, ICMP, ON, Class exercise	3
(check Zeus platform for the date & time)	5	Internet layer protocols (IGP vs EGP, BGP), ACLs, Class exercises	3
(check Zeus platform for the date & time)	6	Application Layer protocols/Security (HTTP/S), Class exercise	3
(check Zeus platform for the date & time)	7	Application Layer protocols/Security (SMTP + PGP), DNS, Class exercises	3
(check Zeus platform for the date & time)	8	NAT, Class exercise, Review/Practice-session	3
Total			24 hours
(check Zeus platform for the date & time)	EXAM		1

GRADING criteria:

Class attendance and responsiveness:	10%
Class assignments (practical work):	40%
Final evaluation (Quiz & Exam):	50%

Lecture 7 Outline

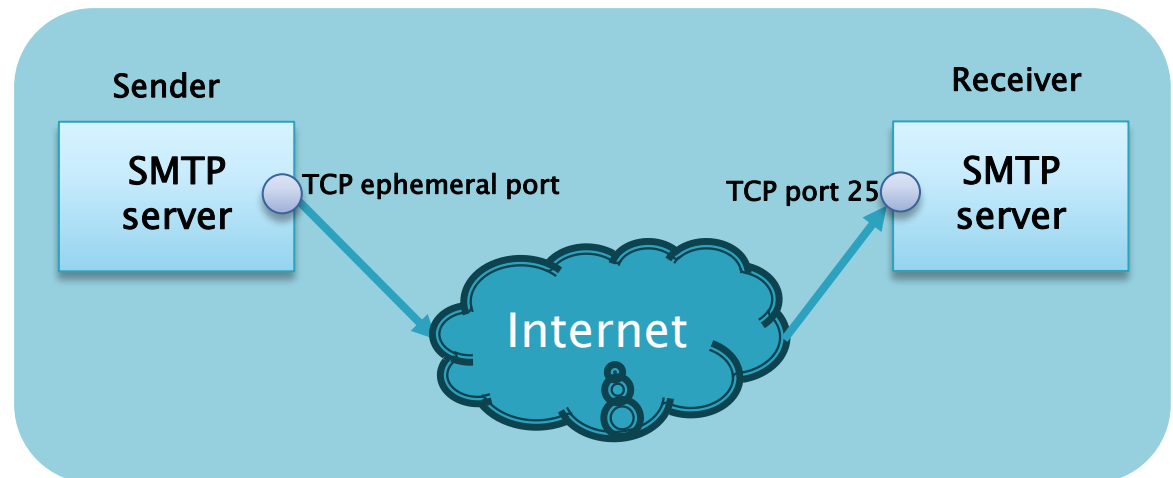
- ▶ **Simple mail transfer protocol (SMTP)**
 - SMTP & email structure
 - Working & limitations
 - PGP, S/MIME & PEM
 - Class exercise 13
- ▶ **Domain Name System (DNS)**
 - Lookup mechanism
 - Caching & updating records
 - Sub-domains & delegation
 - Domain name resolution process
 - Client look-up & cache
 - ICANN & TLDs
 - Registries, Registrars, and Registrants
 - Class exercise 14

Simple mail transfer protocol (SMTP)

(from TCP/IP 5 Layer model perspective)

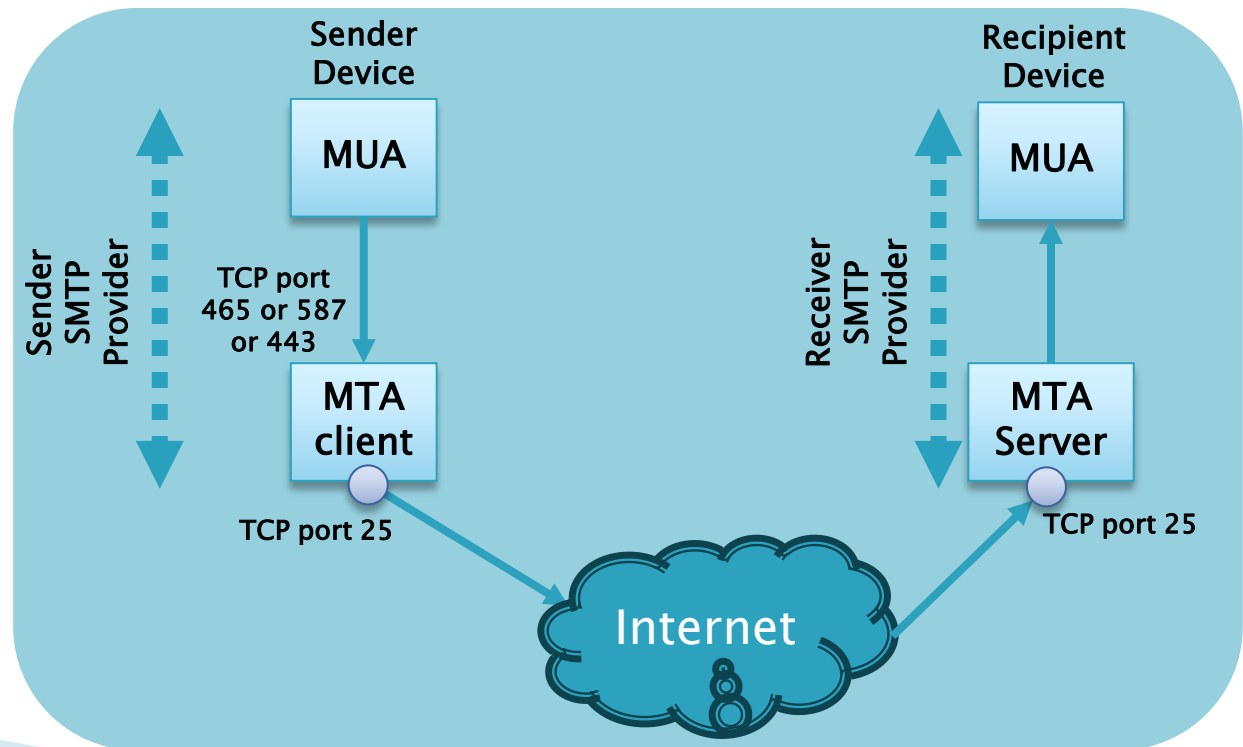
- ▶ First defined in 1982 under RFC 821
 - The goal was to make email sending mechanism simpler
- ▶ When an e-mail is sent from a sender to receiver, in most cases this involves:
 - The sender device sends an email to local SMTP server, which then sends email to recipient's SMTP server

Goal: To transfer mail reliably and efficiently



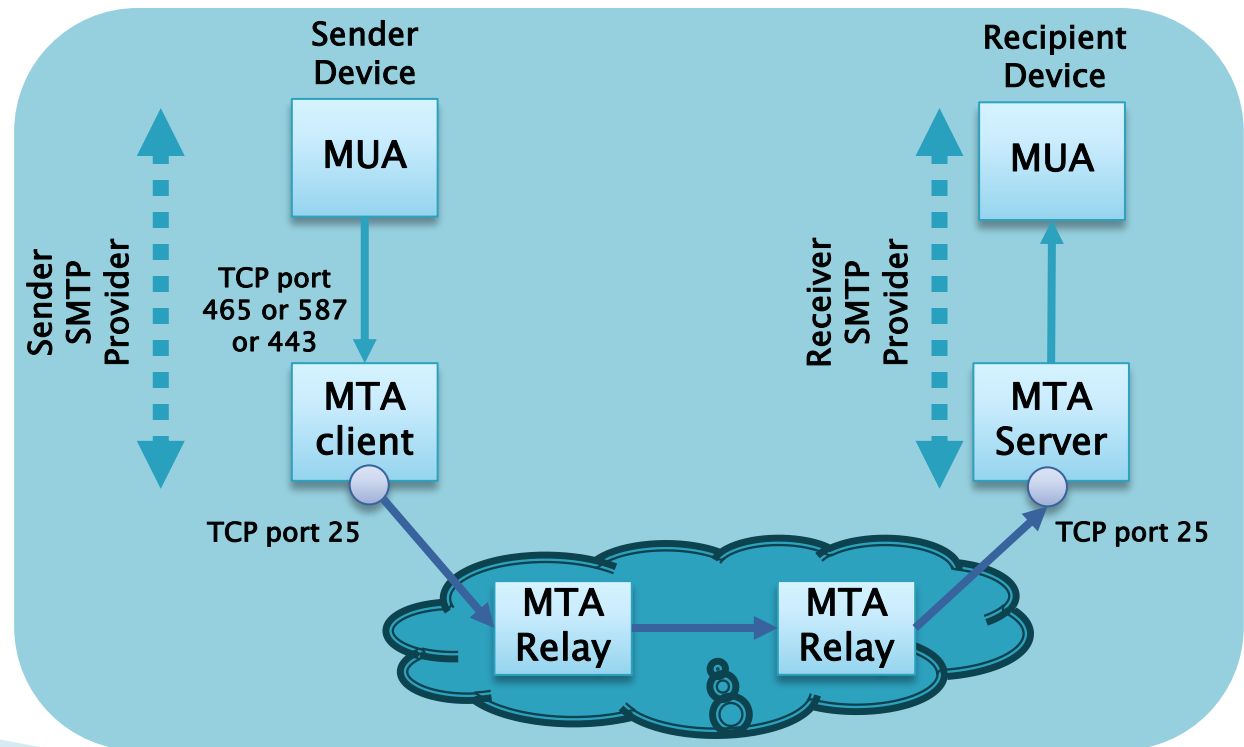
Simple mail transfer protocol (SMTP) – Cont.

- ▶ **SMTP clients and servers** have two main components:
 - Mail User Agent (MUA): Prepares the message and encloses it in an envelope (for e.g., Thunderbird, ...)
 - Mail Transfer Agent (MTA): Perform the actual transfer of email (e.g., Postfix, Exim, Qmail...)



Simple mail transfer protocol (SMTP) – Cont.

- ▶ SMTP also allows the use of **Relays** (a type of MTA) allowing other MTAs to ‘forward’ the email to another MTA for further delivery



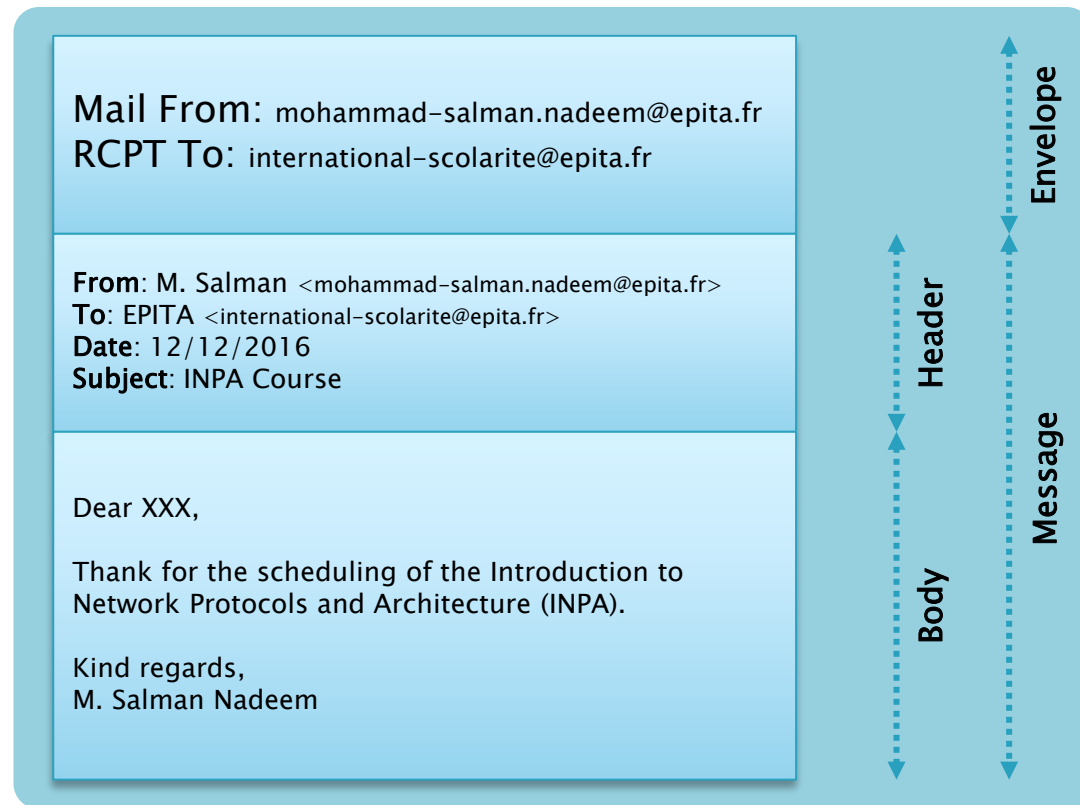
Structure of an email

► Envelope:

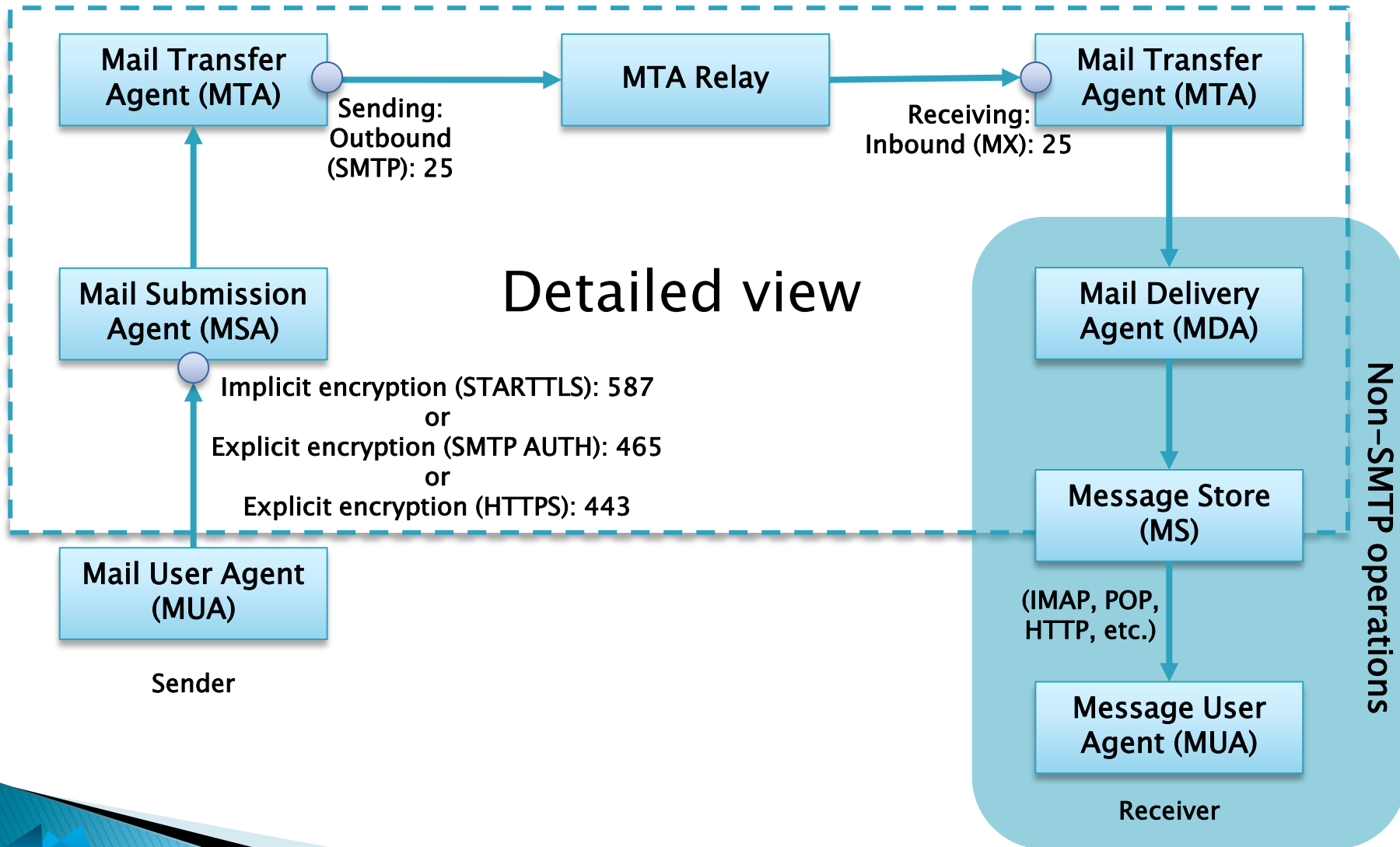
- Sender address
- Receiver address
- Other information

► Message:

- Mail Header: defines the sender, receiver, subject of the message, and some other information
- Mail Body: contains the actual information in the message



Message Handling System (MHS)



SMTP protocol: dialogue keywords

Client Keyword	Arguments
HELO	Sender's Host Domain Name
MAIL FROM:	Email Address of sender
RCPT TO:	Email of Intended recipient
DATA	Body of the message
QUIT	

- ▶ The Server responds with 3 digit code that may be followed by text info
 - 2## – Success
 - 3## – Command can be accepted with more information
 - 4## – Command was rejected, but error condition is temporary
 - 5## – Command rejected, Bad User!
 - ...

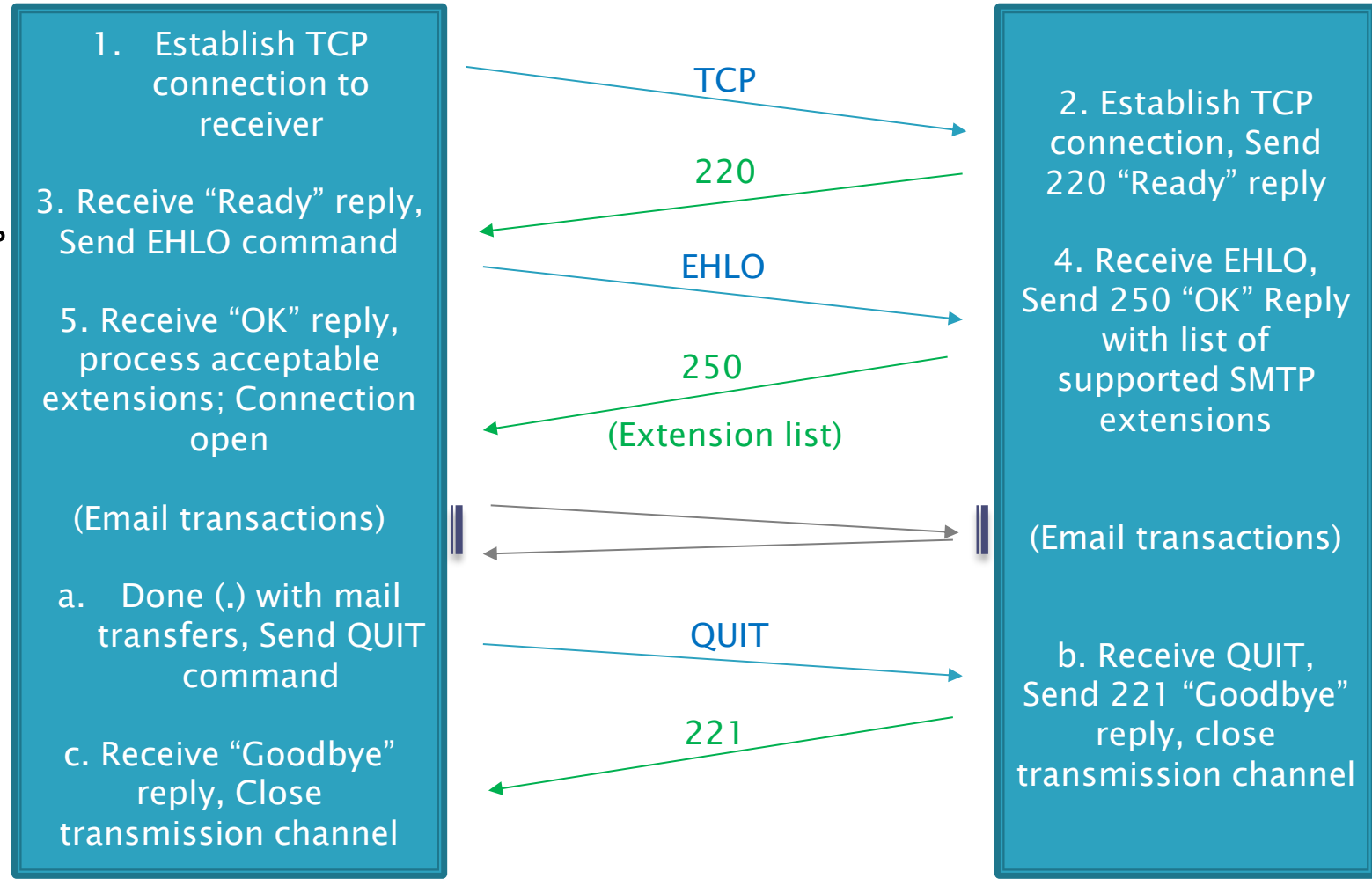
For Enhanced SMTP Codes: www.iana.org/assignments/smtp-enhanced-status-codes/smtp-enhanced-status-codes.xhtml

Message transaction progress

SMTP Sender

SMTP Receiver

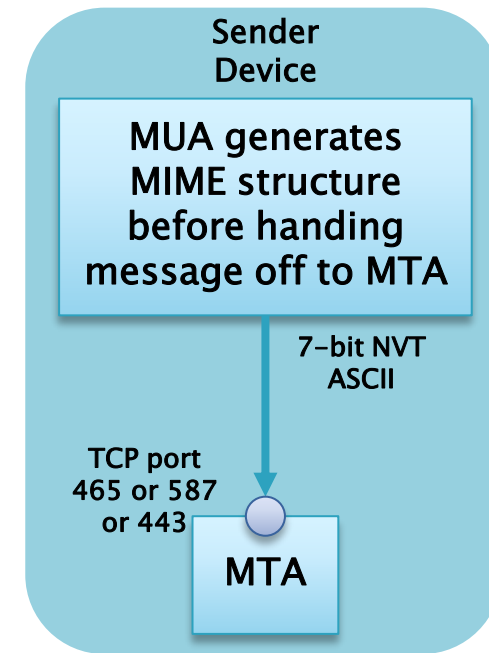
 Extended SMTP
(RFC 2821)



Extensions to SMTP

- ▶ MIME – Multipurpose Internet Mail Extensions
 - Transforms non-ASCII data to NVT (Network Virtual Terminal) ASCII data by encoding it into chunks e.g.,

MIME Type	Subtype	Description
text	plain	Plain text (default format)
text	html	HTML-formatted text (rich text / styling)
multipart	mixed	Combination of text + attachments
application	octet-stream	Generic binary data
image, audio, video	jpeg, png, gif, mpeg, ogg, wav, mp4, x-msvideo	Common image formats, audio files, video files
...	...	...

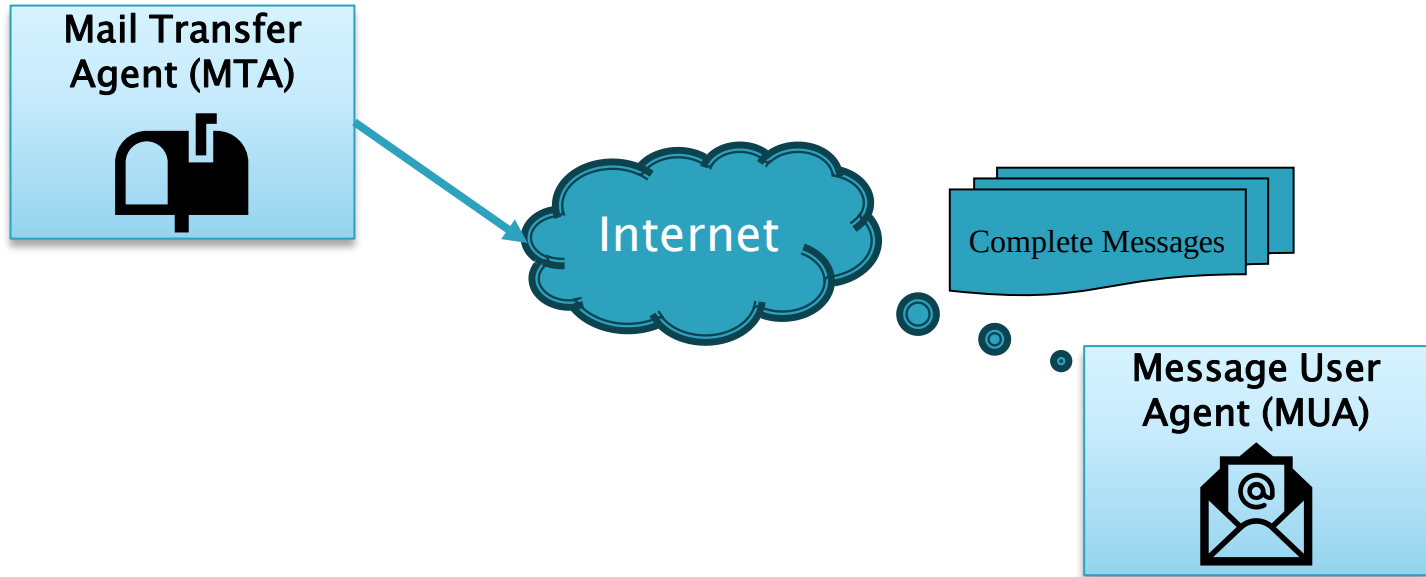


Mail Access Protocols

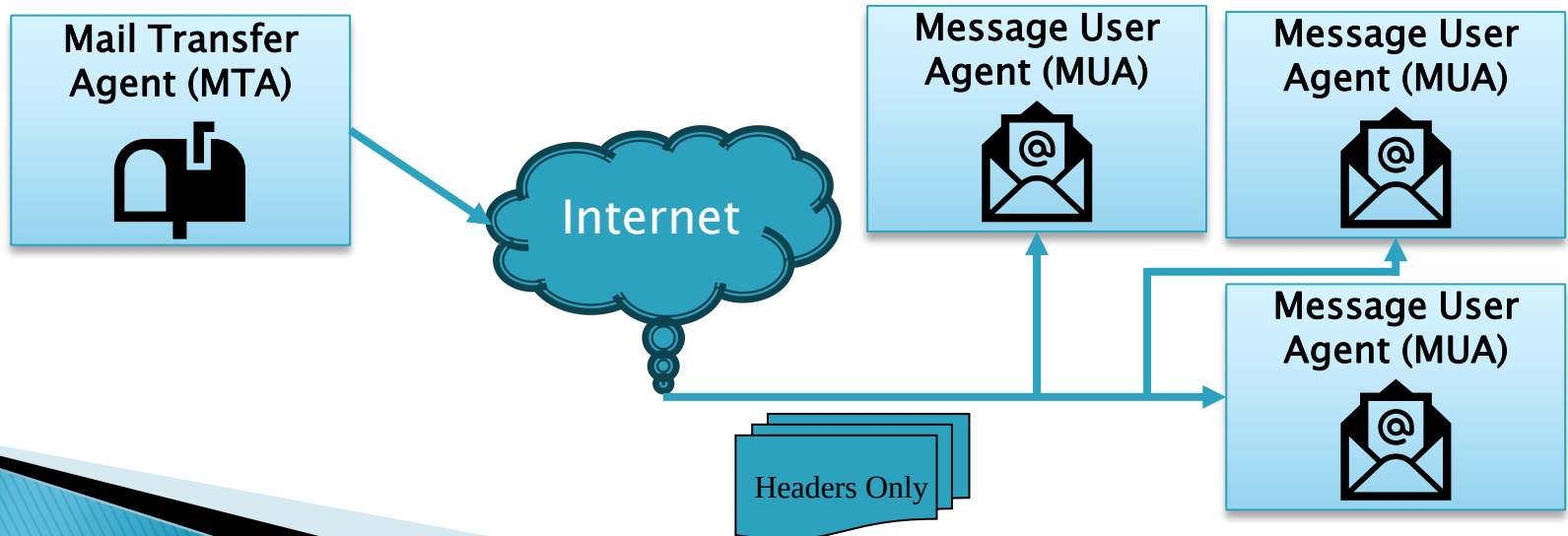
- ▶ The MTA delivers email to the recipients mailbox using Mail Delivery Agent (MDA)
- ▶ The Mail Access Protocols are used by the users to retrieve emails from local SMTP Server (or MDA)
 - Post Office Protocol version 3 (POP3)
 - POP3 (port: 110), POP3-S (port: 995)
 - Internet Message Access Protocol version 4 (IMAP)
 - IMAP4 (port: 143), IMAP4-S (port: 993)

POP vs IMAP

POP3:



IMAP4:



Simple SMTP limitations

- ▶ Only uses NVT (Network Virtual Terminal) 7-bit ASCII format
- ▶ Timeouts problem – If the Client and server have different timeouts, one of them may give up while the other is still busy, unexpectedly terminating the connection
- ▶ No authentication mechanisms
- ▶ Messages are sent un-encrypted
- ▶ Susceptible to misuse (Spamming, faking sender address, ...)
- ▶ ...

PGP, SMIME and PEM

Available tools:

- GnuPG (command-line)
- Mailvelope (browser-extension)
- Mailfence / Protonmail (Webmail based)
- ...

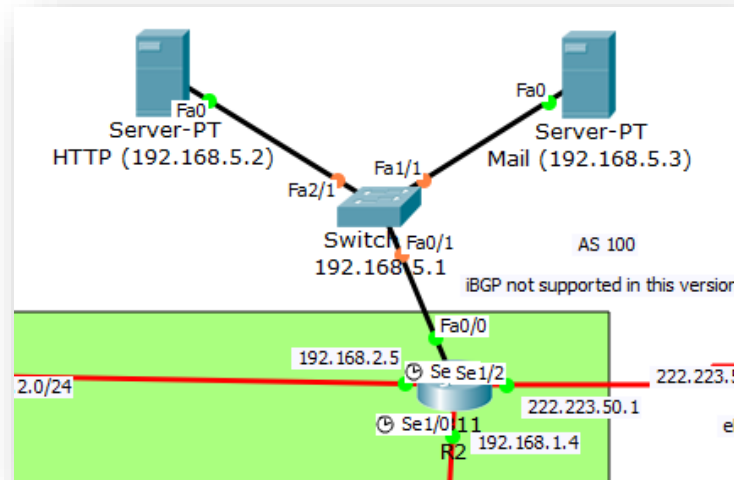
Mechanism	C (Confidentiality)	I (Integrity)	A (Authentication)
PGP: It incorporates mechanisms for authentication, confidentiality, compression, e-mail compatibility and segmentation & reassembly. MAIL & FILE	Symmetric encryption: CAST-128, 3-DES, IDEA	SHA	DSS + SHA or RSA + SHA
S/MIME: S/MIME provides the functionality of Enveloped data, signed data, clear signed data and signed and enveloped data. MIME	Diffie-Hellman (Key Exchange), Triple-DES or RC2/40	SHA-1 / MD5	SHA-1 / MD5 + DSS / RSA
PEM: Mechanism of key management for authentication purposes. Text Based	DES	MD2 / MD5	DES + MD5

Note: SHA1, MD5, DES, 3DES algorithms are now considered as obsolete

Exercise 13: Practical work

► Using your last cisco packet tracer file:

1. On the ASBR router (AS 100), setup an SMTP and HTTP server
2. For SMTP: Configure 2 users i.e., alice and bob
 1. From Bob's device (any device, for instance, in 192.168.4.* network), send an email to Alice's device (any device, for instance, in 192.168.4.* network)
3. Verify using PING and TRACERT/TRACEROUTE, and by visiting the web-site
4. Study the packet transmission in Simulation mode



Save cisco packet tracer file with your 'First_Last-name_Ex-3-13

Deadline: See
'Teams'
Assignment section

Lecture 7 Outline

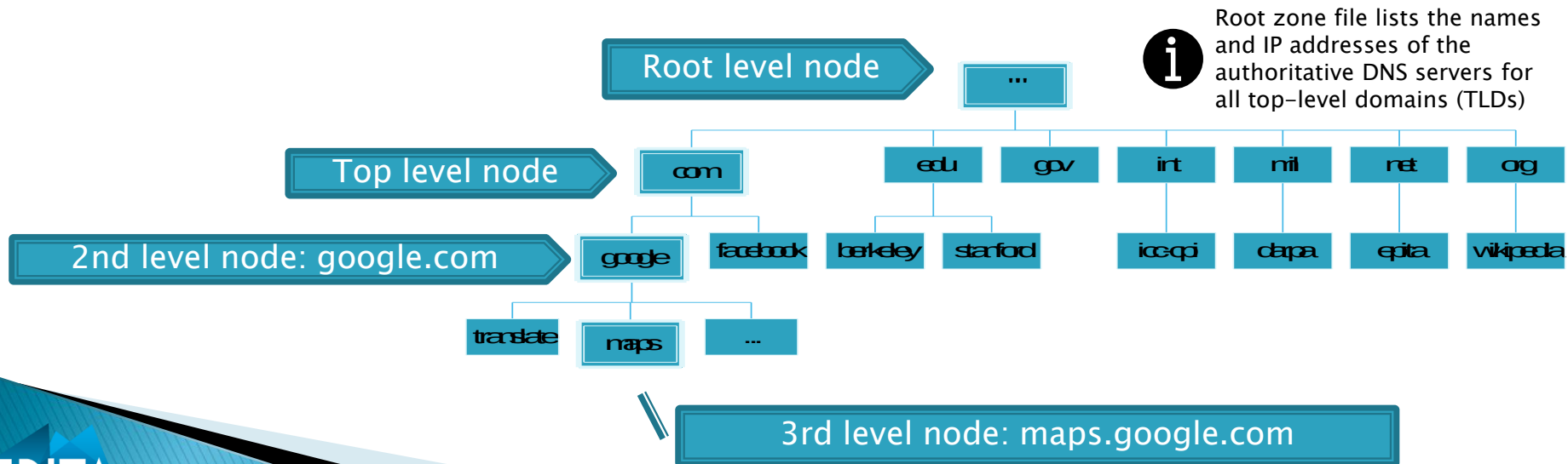
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Domain Name System (DNS)

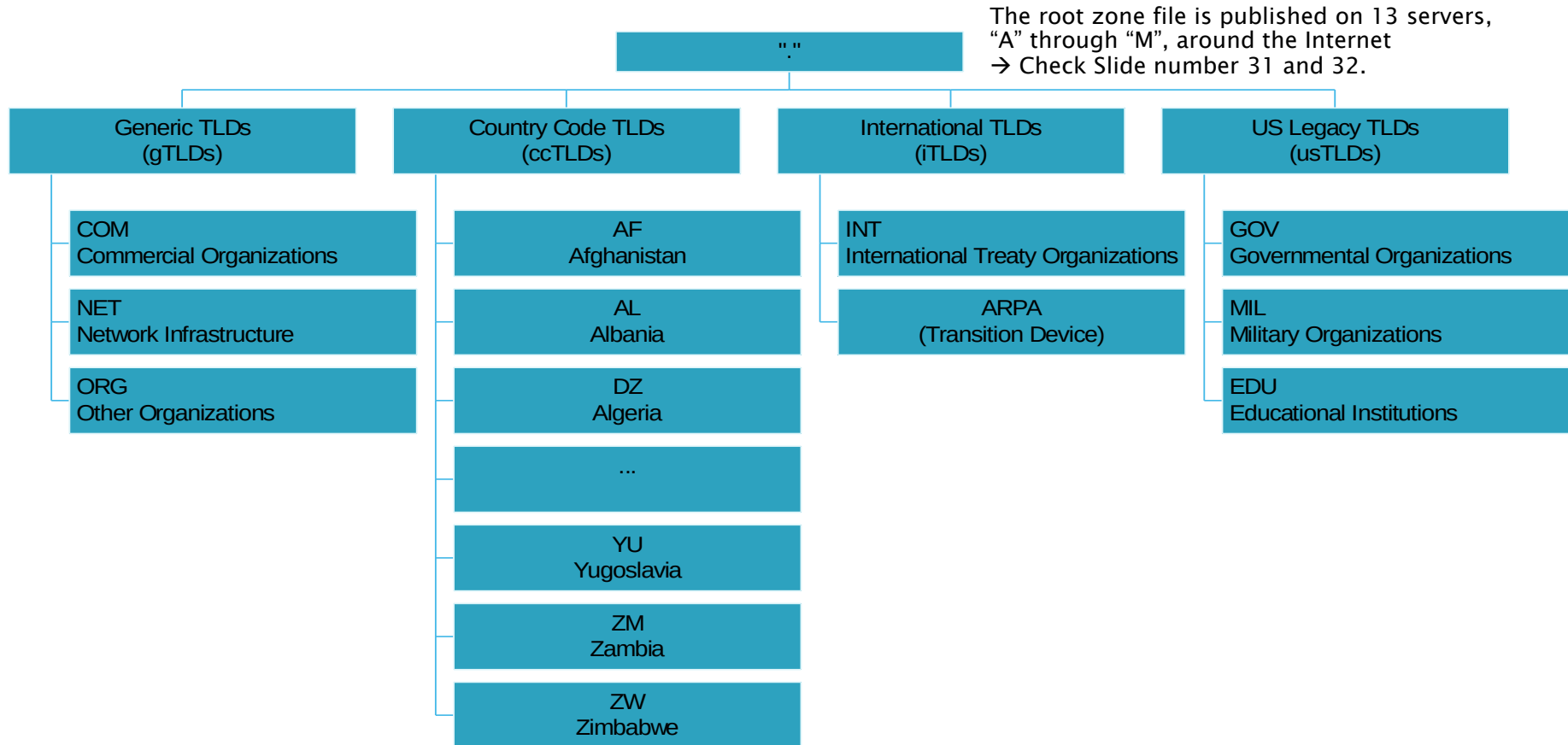
- ▶ In early days of internet, all the information, like name to address mappings, was kept in a single file (e.g., hosts.txt) shared with all the computers connected to the Internet (ARPANET)
 - This became quickly unmaintainable (too much of 'bookkeeping', and was fixed by DNS)
- ▶ Now, we refer to devices/hosts on the Internet by domain names 'epita.fr', or 'Fully Qualified Domain Names' (FQDN) – absolute domain name – if it also includes all labels (e.g., somehost.epita.fr.)
 - Simply because names are easy to remember (than numbers)
 - The mechanism by which they get translated to an IP address and vice versa is handled by **DNS**
 - DNS was specified in RFC 1034 & 1035, and uses the UDP on port 53 (to serve requests)
- ▶ DNS has **three major components**:
 1. Domain Name Space and Resource Records
 2. Name Servers
 3. Resolvers

Domain Name Space

- ▶ Domain Name Space is a (distributed) tree data structure
 - Each node in this tree has a label
 - The root node (written as “”) has an empty label
 - No other label can be empty
 - A sequence of **labels** from a node to the root, separated by dots (‘.’) is read from left to right, and has some limits:
 - 127 levels (root to sub-domains) at max.
 - Each level: 63 chars per label at max.
 - All labels or a domain name must have at max. 253 chars



The Current TLDs



To re-iterate

- ▶ Parts of a domain name:

docs.forge.epita.fr

This country code top-level domain (ccTLD) is 'fr'

The organization is 'epita'

The sub-domain is called 'forge'

The specific 'label' for documentation is 'docs'

→ A domain is a group of labels

Resource Records (RR)

- ▶ Each node in Domain Name Space can keep information in the form of resource record sets (RRset) containing:
 - Owner: of this record (domain name of the node which keeps this information)
 - Type: a 16-bit value, specifying what this resource record is about (e.g., SOA – identifies the start of a zone, NS – DNS server, A – IPv4, AAAA – IPv6)
 - Class: a 16-bit value, identifying a protocol (e.g., IN – internet, ANY – used only in queries)
 - TTL: a 32-bit Time-To-Live value, in units of seconds, indicating how long this resource record can be cached
 - rdata: data of the resource (depends on the resource type)
- ▶ All resource records in a resource record set has the same name, type and class

Name servers (NS)

- ▶ Name servers (e.g., BIND) are server programs, which hold some information about the Domain Name Space
 - Each name server may have complete and authoritative information only about a small part of the Domain Name Space, and possibly cache some other data in the Domain Name Space
- ▶ Data is maintained locally, but retrievable globally
 - No single device has all DNS data – and DNS lookups can be performed by any device
- ▶ Act as a distributed directory service
 - Resources names $\leftrightarrow$ respective addresses

Name servers (NS): Bind

```
msn-box@msnbox-vm:~$ systemctl status bind9
● named.service - BIND Domain Name Server
   Loaded: loaded (/lib/systemd/system/named.service; enabled; vendor preset: enabled)
   Active: active (running) since Sat 2020-12-12 14:27:16 CET; 21s ago
     Docs: man:named(8)
    Main PID: 2194 (named)
      Tasks: 5 (limit: 4650)
     Memory: 27.5M
    CGroup: /system.slice/named.service
            └─2194 /usr/sbin/named -f -u bind

Dez 12 14:27:17 msnbox-vm named[2194]: network unreachable resolving './DNSKEY'
Dez 12 14:27:17 msnbox-vm named[2194]: network unreachable resolving './NS/IN'
Dez 12 14:27:17 msnbox-vm named[2194]: network unreachable resolving './DNSKEY'
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Dez 12 14:27:17 msnbox-vm named[2194]: network unreachable resolving './NS/IN'
Dez 12 14:27:17 msnbox-vm named[2194]: managed-keys-zone: Initializing automatic key generation
Dez 12 14:27:17 msnbox-vm named[2194]: resolver priming query complete
```

Screenshot from: systemctl output displaying BIND9 service status on Ubuntu Linux (BIND version 9). Captured from locally running instance, December 2020.

```
options {
    directory "/var/cache/bind";

    // If there is a firewall between you and nameservers you want
    // to talk to, you may need to fix the firewall to allow multiple
    // ports to talk.  See http://www.kb.cert.org/vuls/id/800113

    // If your ISP provided one or more IP addresses for stable
    // nameservers, you probably want to use them as forwarders.
    // Uncomment the following block, and insert the addresses replacing
    // the all-0's placeholder.

    // forwarders {
    //     0.0.0.0;
    // };

    //=====
    // If BIND logs error messages about the root key being expired,
    // you will need to update your keys.  See https://www.isc.org/bind-key
    //=====

    dnssec-validation auto;

    listen-on-v6 { any; };
};

<etc/bind/named.conf.options" [readonly] 24L, 846C      1,1      All
```

Screenshot from: BIND configuration file (/etc/bind/named.conf.options) opened in terminal text editor on Ubuntu. Captured from locally configured system, December 2020.

Bind acts as an authoritative name server for domains, and as well as a recursive resolver in the network

Resolvers

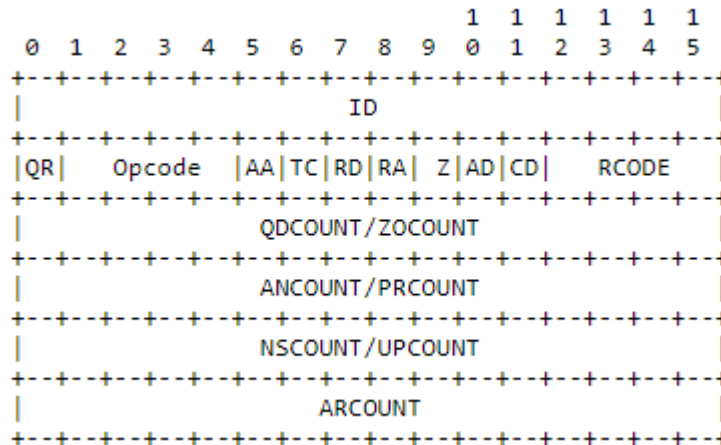
- ▶ Programs in user device
 - Send queries to name servers (NS), to extract information from Domain Name Space → this process is called DNS lookup
 - E.g., dig: a DNS lookup utility
- ▶ Iterative vs. Recursive Queries
 - Iterative approach: where the resolver repeatedly queries different servers until it finds the answer
 - Recursive approach (common one): where the resolver sends the query only to a single server, then that server repeatedly queries different name servers until it finds and returns the answer to the resolver
- ▶ If a DNS response is large, DNS lookup can be retried over TCP

DNS Header section & format

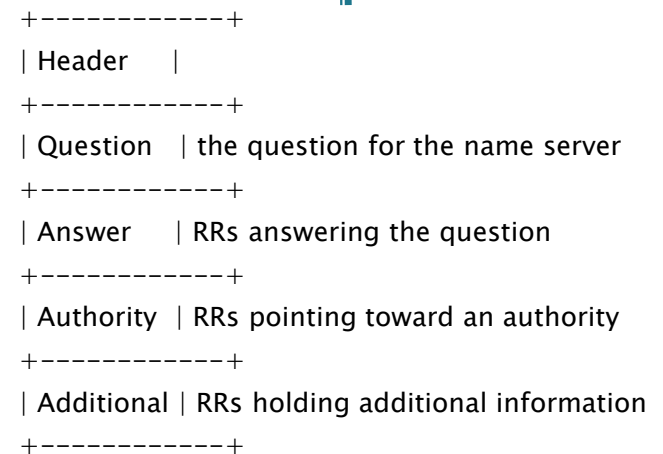
A resolved domain name is cached by NS:

- It expires (under the defined TTL: Time To Live)
- TLD's typically get cached in local name server
- Root name servers are not often visited

Details of the Header section



DNS message always has the same format, it does not matter if it is a query or response



Source: <https://tools.ietf.org/html/rfc2929>

Resolution process (1 / 2)

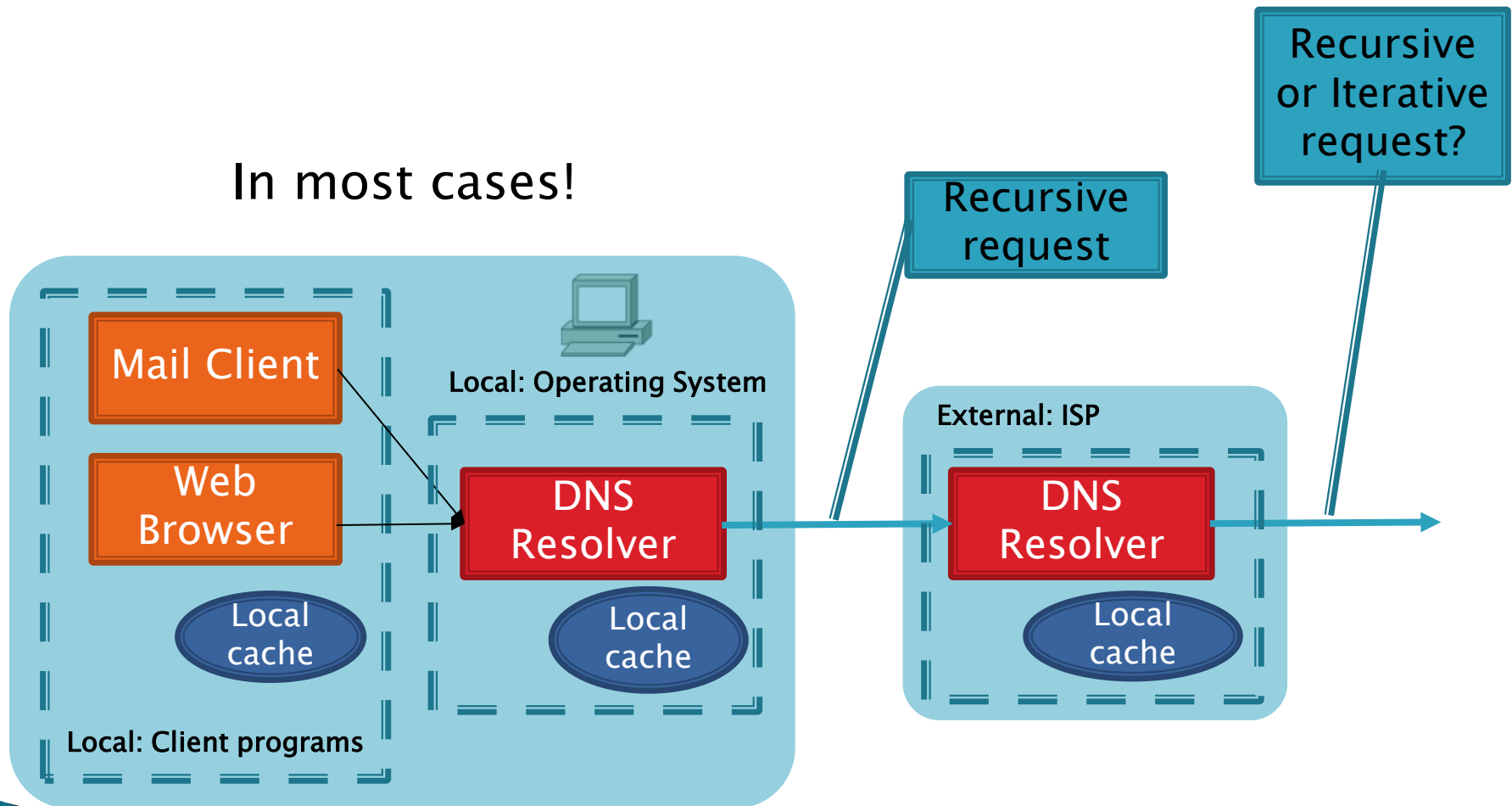
- ▶ Take 'dig' DNS look-up utility as an example
 - The query is sent to the system default name server (e.g., on GNU/Linux -> /etc/resolv.conf)
 - Only answer is displayed, query is hidden by default
- ▶ Query any domain using private or public DNS, and enable the display of query request
 - `dig @8.8.8.8 epita.fr IN A +qr`
- ▶ Lets see the effect of Recursion Desired (RD) by not setting it with `+nordflag`:
 - `dig @8.8.8.8 epita.fr IN A +qr +nordflag`
 - Because Google's Public DNS (8.8.8.8) is not authoritative for epita.fr, when no recursion is set, it simply returns nothing (or a SERVFAIL)

Resolution process (2/2)

- ▶ Find the authoritative name server of a given domain name:
 - Let us start from the root domain:
 1. `dig //root name servers`
 2. `dig @a.root-servers.net fr NS +qr +nordflag`
//name servers of 'com' -> authoritative
 3. `dig @e.ext.nic.fr epita.fr NS +qr +nordflag`
//name server of 'epita.fr'
 4. `Dig @banjo.ionis-it.com epita.fr A +qr +nordflag`
//Using authoritative name servers of epita.fr to ask resource records: A (IPv4)
 5. ...

Client lookup & Local cache (2 / 2)

In most cases!



Internet Corporation for Assigned Names and Numbers (ICANN)

- ▶ ICANN's role in DNS hierarchy: to oversee the management of Internet resources including
 - Addresses
 - Delegating blocks of addresses to the regional registries
 - Protocol identifiers and parameters
 - Allocating port numbers, etc
 - Names
 - Administration of the root zone file
 - Oversee the operation of the root name servers
- ▶ Connecting to the Internet implies use of the existing DNS hierarchy – that's the rule!

Root servers

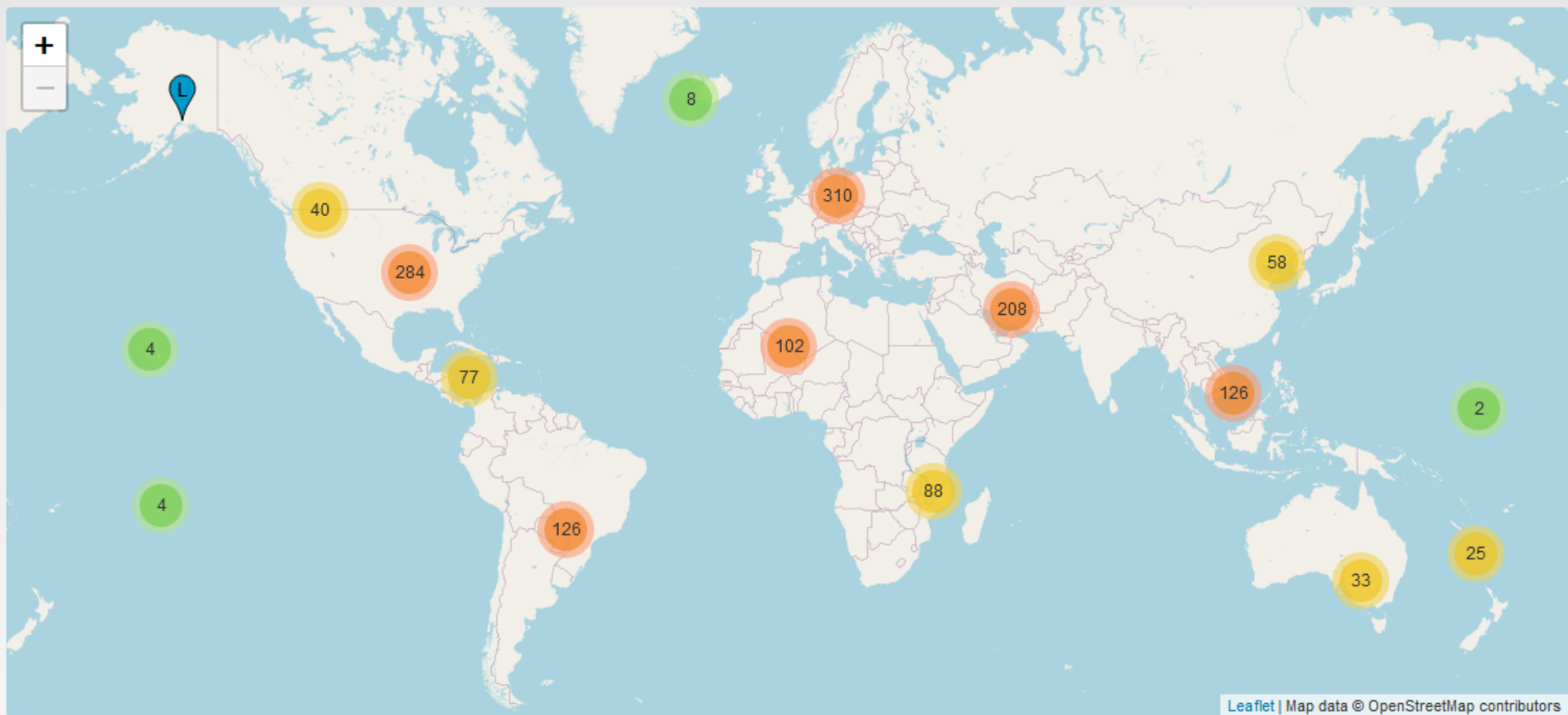
- ▶ The authoritative name servers, that serve the root zone file, are configured in the DNS root zone as 13 named authorities (“A” through “M”):

List of Root Servers

HOSTNAME	IP ADDRESSES	OPERATOR
a.root-servers.net	198.41.0.4, 2001:503:ba3e::2:30	Verisign, Inc.
b.root-servers.net	199.9.14.201, 2001:500:200::b	University of Southern California, Information Sciences Institute
c.root-servers.net	192.33.4.12, 2001:500:2::c	Cogent Communications
d.root-servers.net	199.7.91.13, 2001:500:2d::d	University of Maryland
e.root-servers.net	192.203.230.10, 2001:500:a8::e	NASA (Ames Research Center)
f.root-servers.net	192.5.5.241, 2001:500:2f::f	Internet Systems Consortium, Inc.
g.root-servers.net	192.112.36.4, 2001:500:12::d0d	US Department of Defense (NIC)
h.root-servers.net	198.97.190.53, 2001:500:1::53	US Army (Research Lab)
i.root-servers.net	192.36.148.17, 2001:7fe::53	Netnod
j.root-servers.net	192.58.128.30, 2001:503:c27::2:30	Verisign, Inc.
k.root-servers.net	193.0.14.129, 2001:7fd::1	RIPE NCC
l.root-servers.net	199.7.83.42, 2001:500:9f::42	ICANN
m.root-servers.net	202.12.27.33, 2001:dc3::35	WIDE Project

Screenshot from: Internet Assigned Numbers Authority (IANA), List of Root Servers. Retrieved from <https://www.iana.org/domains/root/servers>

“Root servers” are a network of servers



As of 2023-06-25T11:34:58Z, the root server system consists of 1708 instances operated by the 12 independent root server operators.

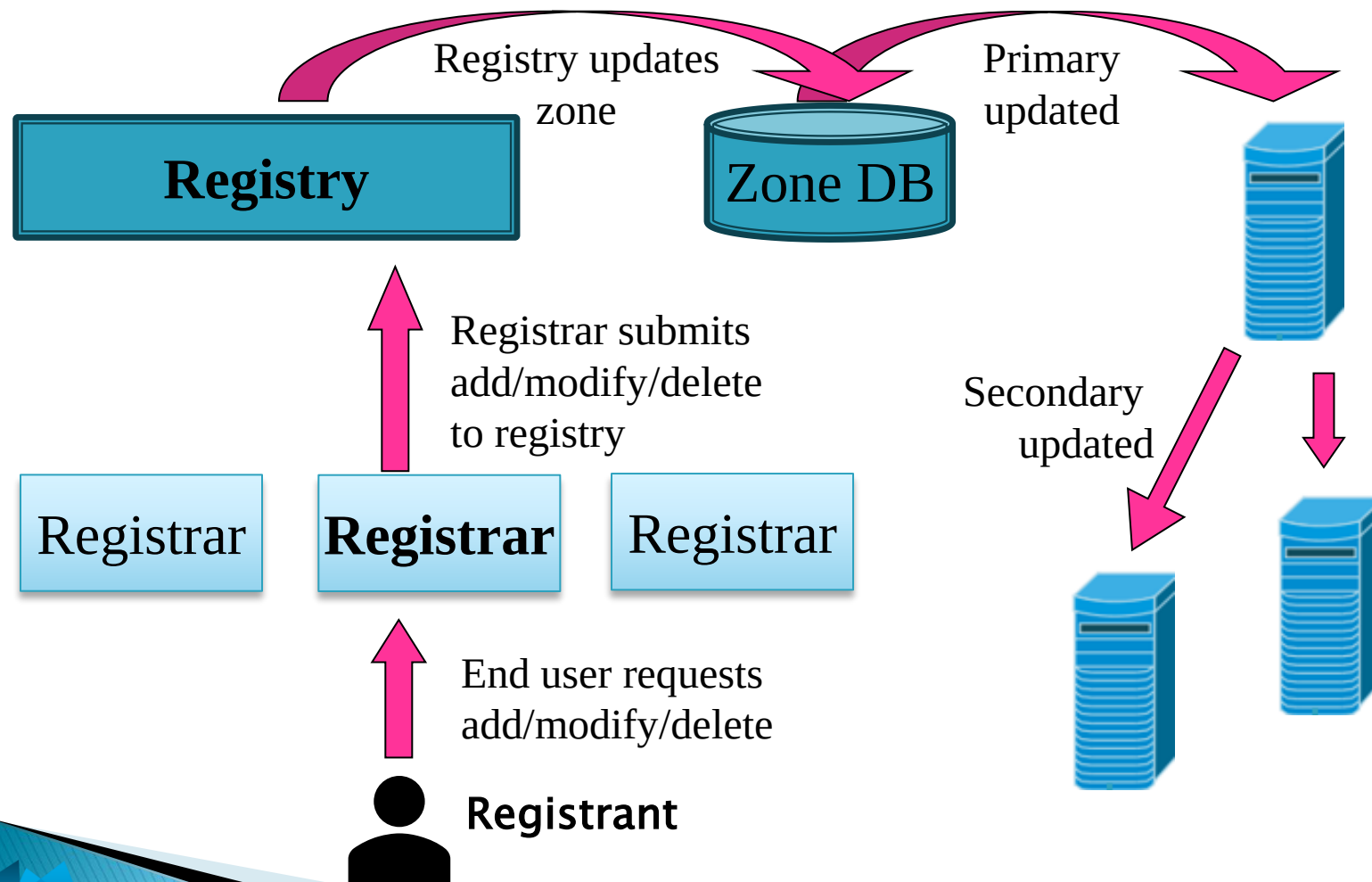
Screenshot from: Root Server System map showing global distribution of root server instances. Retrieved from <https://root-servers.org>

Registries, Registrars, and Registrants

A classification of roles in the operation of a domain name space

- ▶ Registry
 - The name space's database
 - The organization which has edit control of that database and runs the authoritative name servers for that name space
- ▶ Registrar
 - The agent which submits change requests to the registry on behalf of the registrant
- ▶ Registrant
 - The entity which makes use of the domain name

Registries, Registrars, and Registrants



Exercise 14a: Practical work

- ▶ Perform DNS queries and study the responses, using following tools:

→ dig

```
dig epita.net
```

```
dig @1.1.1.1 epita.net //1.1.1.1 is the specified name server
```

```
dig MX epita.net //Replace MX with other DNS record types e.g., A, CNAME, AAAA, NS, ...
```

→ nslookup

```
nslookup epita.net
```

```
nslookup epita.net 1.1.1.1 //1.1.1.1 is the specified name server
```

```
nslookup -debug epita.net //-d2 for more verbose answer
```

→ host

```
host -a epita.fr
```



Direct 'traceroute' to DNS server will likely not succeed, use 'traceroute -U X.X.X.X' instead

*Save your txt/doc
file with your
'First_Last name'*

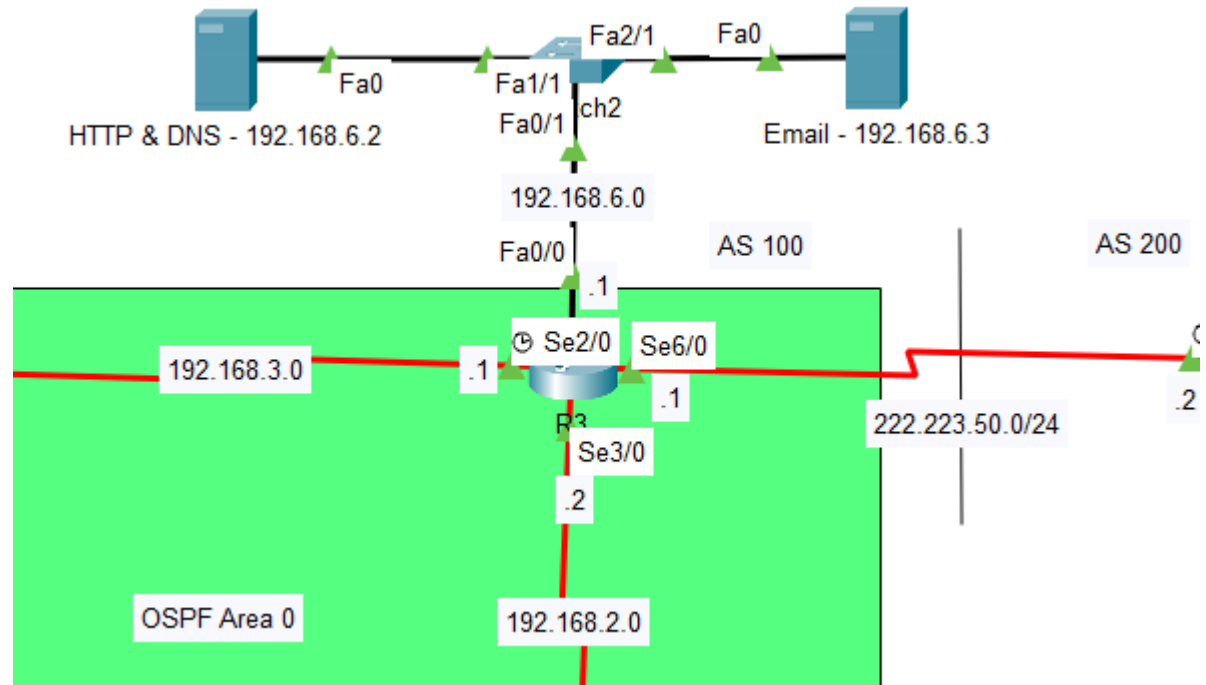
**Deadline: See
'Teams'
Assignment section**

Exercise 14b: Practical work

- ▶ Using your last cisco packter tracer file:
 1. Configure DNS service on the HTTP server – at ASBR router (AS 100)
 2. Configure the DNS server on all end-devices

Deadline: See
'Teams'
Assignment section

*Save cisco packet
tracer file with
your 'First_Last-
name_Ex-3-14*



Lecture 7 ends here

- ▶ **Course Slides: Go to MS Teams:**
Introduction to Network protocols and architecture
→ Files section
- ▶ **Send your questions by email**
OR via direct message using MS Teams
- ▶ **Thank You!**